

Jonah Wood

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EDUCATION

Northeastern University

B.S. in Electrical and Computer Engineering

Boston, MA

Expected May 2030

Salt Lake Community College

Associate of Science

Salt Lake City, UT

Apr. 2026

Beehive Science and Technology Academy

Honors Diploma

Sandy, UT

Apr. 2026

EXPERIENCE

Engineering Intern

Jul. 2026 – Sep. 2026

Yale Cordage

Saco, ME

- Designed, built, budgeted, ordered, and programmed a patent-pending “Smart Reeve Plate”: an end-to-end system that guides twisting-machine operators through rope reeving patterns via a scan-to-LED workflow.
- Architected the full pipeline in Node-RED: barcode work-order scan → product-config lookup → PLC → HMI selection → pattern retrieval → Arduino-driven LED display on a custom-machined plate.
- Designed a human-readable topology language and custom parser that compiles reeve patterns into hole-position and end-count data, plus authoring and editing tools built around documentation and ease of use.
- Engineered a custom hex encoding to fit pattern data within limited PLC memory, and designed the custom PCB and LED layout that renders each pattern on the plate.
- Parsed 3,000+ rows of unstructured product data into a structured database using AI tools, saving the company tens of thousands of dollars and weeks of manual effort.

Lead Designer & Build Team Lead

2023 – 2026

Hive 3747 – FIRST Tech Challenge Robotics

Sandy, UT

- Led the build team to back-to-back Utah State Championship Inspire Awards (1st place) and #1 robot-game rankings, qualifying for the FIRST World Championship all three seasons.
- Designed and fabricated virtually every robot mechanism across three years; owned CAD in Fusion 360 and coordinated closely with the programming team.

PROJECTS

cpp-game-engine | C++20

- 2D game engine built from scratch with SDL2/OpenGL: fixed-timestep (240 Hz) physics decoupled from rendering, SAT + AABB collision resolution, a composition-based entity/controller system, and levels authored as PNG images mapped to game objects.

Evolution Engine | Go

- Language-agnostic genetic-algorithm optimizer that evolves weight vectors for any program over a stdin/stdout pipe; cut per-generation time ~900x (900 ms → <1 ms) via a persistent worker-process pool with channel/WaitGroup concurrency and adaptive mutation scaling.

canvas-achieve-sync | Python

- Consolidates Canvas assignments, Achieve textbook work, and the Northeastern academic calendar into a single Google Tasks list.

neu-clubs | Python

- Scrapes Northeastern’s Engage directory into a searchable table of all 933 student groups, with descriptions and Instagram handles.

TECHNICAL SKILLS

Languages: Java, C++, C (Embedded), Go, Python

Tools & Fabrication: Fusion 360 (CAD), 3D Printing, Laser Cutting, PCB Design, Node-RED, PLC/HMI, Arduino, Git

HONORS & AWARDS

National Merit Scholarship Finalist · Congressional Award Gold Medal

AP Scholar with Distinction — 8 AP exams (seven 5s, one 4), including Physics C: Mechanics, Calculus AB, and Computer Science A